

Homework (Habanero)

- Q1.** What does the interquartile range (IQR) represent in a box-and-whisker plot?
- (A) The range of the data set
(B) The median of the data set
(C) The difference between Q_3 and Q_1
(D) The range of the middle 50
(E) The standard deviation of the data
- Q2.** A box-and-whisker plot shows a dataset with $Q_1 = 12$, $Q_3 = 20$, and a median of 16. What is the interquartile range (IQR)?
- (A) 4
(B) 8
(C) 12
(D) 16
(E) 20
- Q3.** Which of the following statements about box-and-whisker plots is FALSE?
- (A) The median is always at the center of the box
(B) Outliers are always below Q_1 or above Q_3
(C) The interquartile range (IQR) is always positive
(D) A box-and-whisker plot can be used to compare multiple datasets
(E) The box represents the range of the data
- Q4.** In a study on the dosage of a new medicine, the box-and-whisker plot shows an outlier at $25mg$. If $Q_1 = 10mg$ and $Q_3 = 18mg$, what can be concluded about the outlier?
- (A) It is a valid data point
(B) It is an error in data collection
(C) It is more than $1.5 \times IQR$ away from Q_3
(D) It is less than $1.5 \times IQR$ away from Q_3
(E) It is the median of the dataset
- Q5.** A dataset of cell growth rates has a box-and-whisker plot with $Q_1 = 2 \frac{mm}{day}$, $Q_3 = 4 \frac{mm}{day}$, and a median of $3 \frac{mm}{day}$. What is the interquartile range (IQR)?
- (A) $1 \frac{mm}{day}$
(B) $2 \frac{mm}{day}$
(C) $3 \frac{mm}{day}$
(D) $4 \frac{mm}{day}$
(E) $5 \frac{mm}{day}$
- Q6.** Which of the following box-and-whisker plots represents a dataset with the largest interquartile range (IQR)?
- (A) $Q_1 = 5$, $Q_3 = 15$
(B) $Q_1 = 10$, $Q_3 = 20$
(C) $Q_1 = 8$, $Q_3 = 12$
(D) $Q_1 = 2$, $Q_3 = 10$
(E) $Q_1 = 1$, $Q_3 = 5$
- Q7.** A box-and-whisker plot shows the distribution of medicine dosages with $Q_1 = 5mg$, $Q_3 = 15mg$, and an outlier at $30mg$. What can be concluded about the outlier?
- (A) It is within $1.5 \times IQR$ of Q_3
(B) It is more than $1.5 \times IQR$ away from Q_3
(C) It is the median of the dataset
(D) It is Q_1
(E) It is Q_3
- Q8.** Which of the following statements about outliers in a box-and-whisker plot is TRUE?
- (A) Outliers are always below Q_1
(B) Outliers are always above Q_3
(C) Outliers can be below Q_1 or above Q_3
(D) Outliers are never below Q_1 or above Q_3
(E) Outliers are the median of the dataset

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset: 12, 15, 18, 20, 22, 25, 30, 35, 40. Identify the quartiles, interquartile range (IQR), and any outliers.
- Q10.** A study on cell growth rates collected the following data: $2 \frac{mm}{day}$, $2.5 \frac{mm}{day}$, $3 \frac{mm}{day}$, $3.5 \frac{mm}{day}$, $4 \frac{mm}{day}$, $4.5 \frac{mm}{day}$, $10 \frac{mm}{day}$. Create a box-and-whisker plot and determine if there are any outliers. Justify your answer.

Chef's Tips